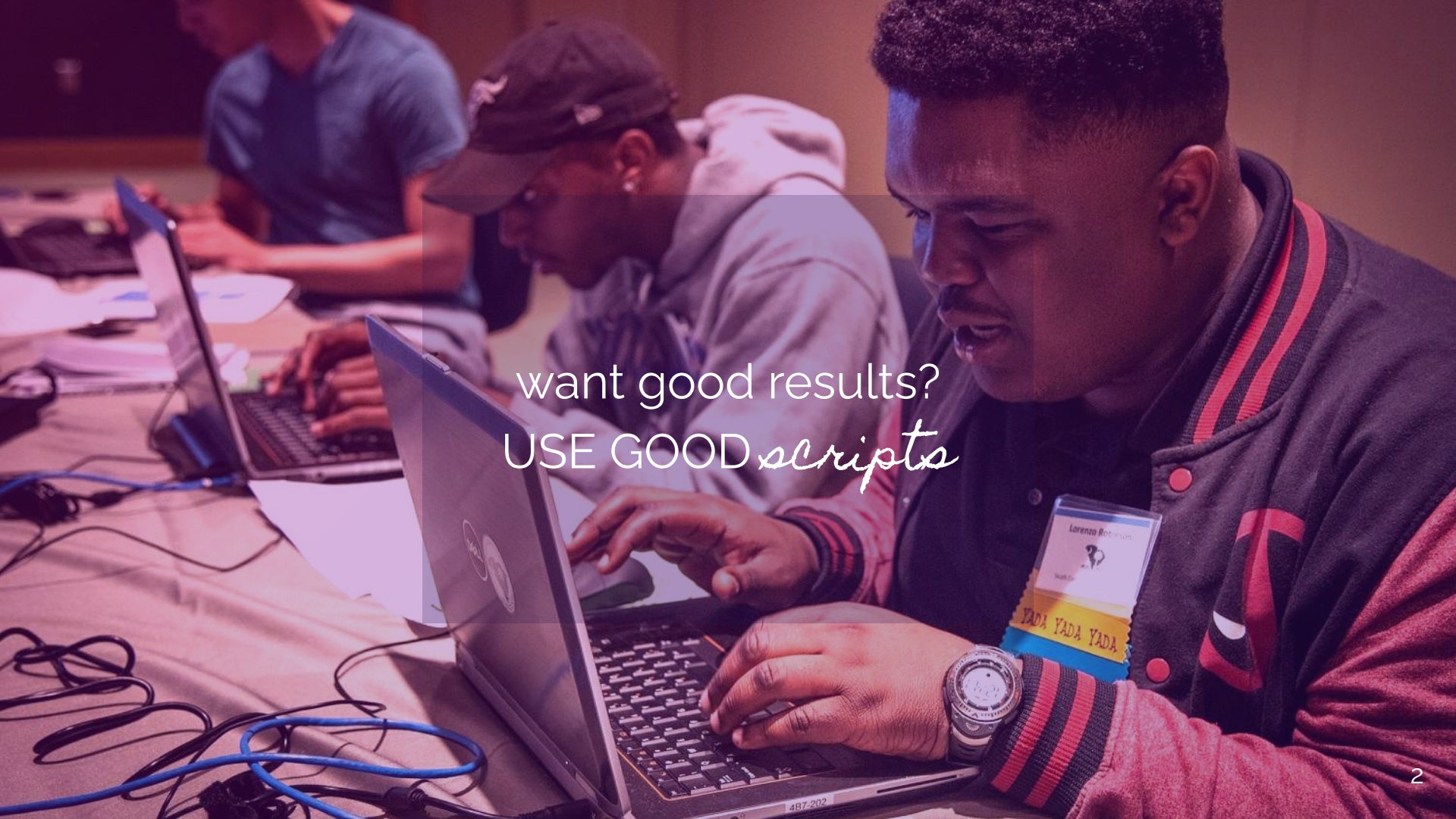
6.2 BASICS and *batch*

By Clement Chan


want good results?
USE GOOD scripts
YADA YADA YADA



PROGRAMMING BASICS

An AP Computer Science A crash course



Programming is taking a common approach and refining it - its speed, its flexibility, its effectiveness - to produce a work of art.

-Clement Chan

- Each line (statement) in the file is executed in order



```
print “Hello, world!”
a = 1
print a
```

- Above would print:

```
Hello, world!
1
```

Programming *Variables* and *Comments*

- Variables are things that can change
  - Marked in a special format, depending on language
- Comments are not executed
  - Usually designated by “#” or “//”
  - Used for documentation

```
$user = Jane
net user $user Passw0rd!123 /add #adds Jane
$user = Joe
net user $user Passw0rd!123 /add #adds Joe
```

Programming *Controls*

- Statements can be **controlled** to execute conditionally or repeat

```
if (firewall off) {
    turn on firewall
}

for (user in userlist) {
    set password
}
```

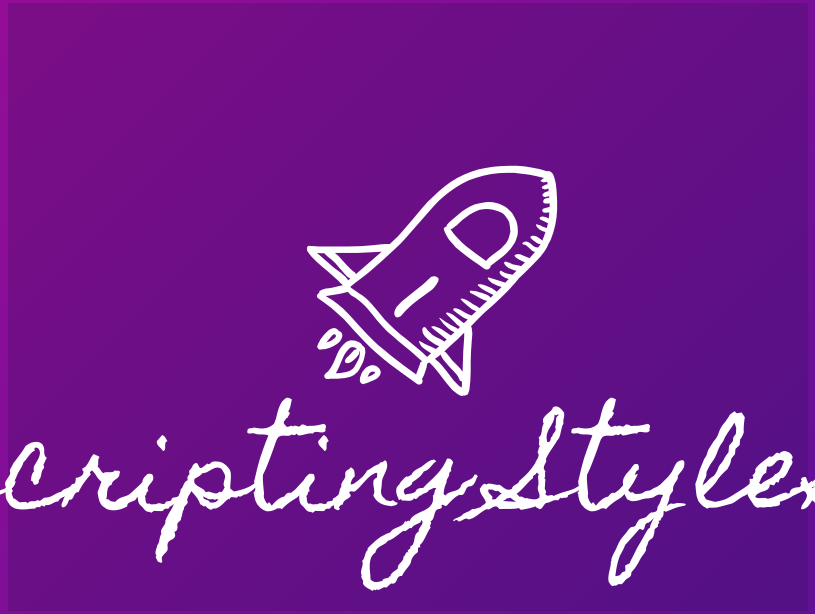
```
while (points is not 100) {
    gain points
}
```

How would you make a users script? Use pseudocode.

Challenge yourself to think like a programmer!

Programming *User example*

```
// prior to, make a “myusers” and a “readmeusers” file
// this block of code configures users and their properties
for (user in myusers) {
        if (user not in readmeusers) {
                disable user
        } else {
                enable user
                set password
                set password expires
        }
}
```



You can program in different fashions

Unique syntax

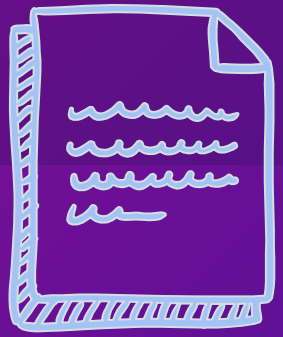
System.out.print(“Hi”);
print “Hi”
print(“Hi”)
printf(“Hi”);
echo Hi
echo “Hi”;
Write-Host Hi

Unique functionalities

Batch and Powershell can use direct commands

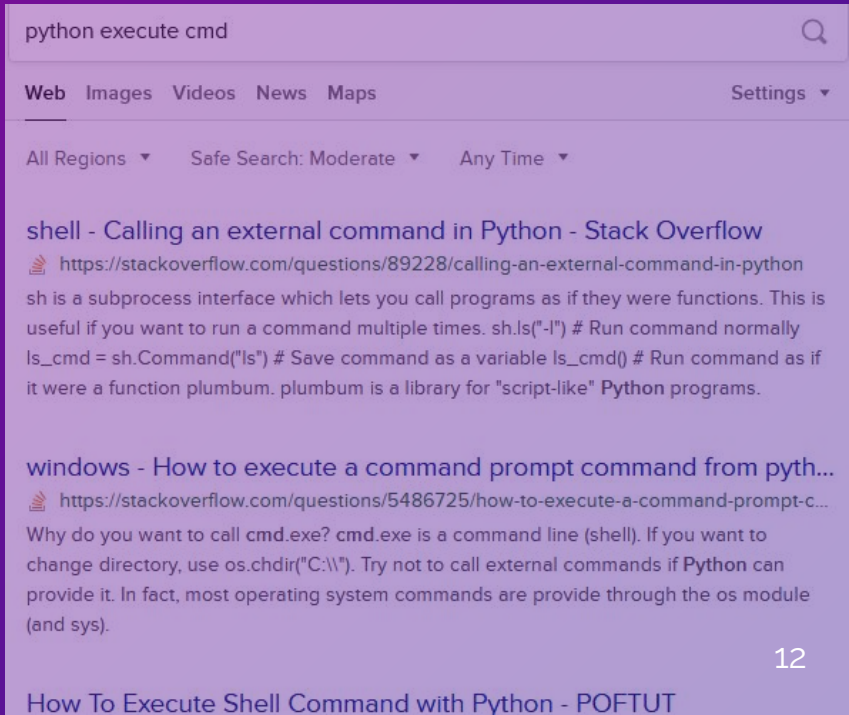
Java and Python are more general - need **libraries**

Libraries

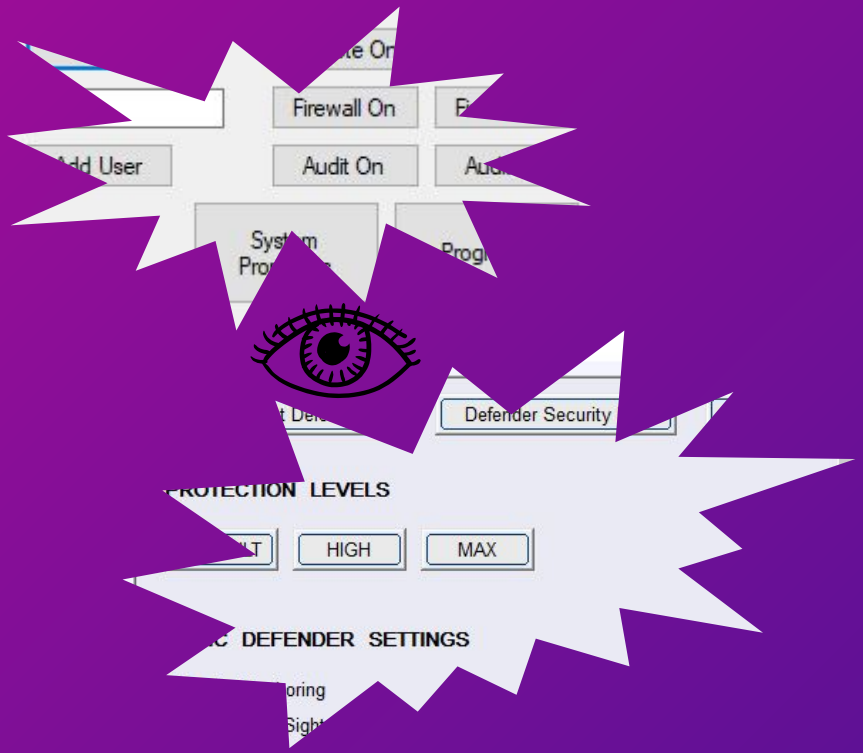


Import libraries for more functions to use

Search online for correct usage



A GUI IS *worth* A THOUSAND LINES



GUIs are difficult to make, but they are effective for quick access or conditional security practices

Batch Crash Course

Syntax in batch

Print a message

You can print variables or environment variables

- variables - temporary
- environment variables - global

echo Hello there!

echo %username%

```
C:\Users\Clement>echo %username%
Clement
```

```
C:\Users\Clement\Desktop>echo Hello there!
Hello there!

C:\Users\Clement\Desktop>echo Clement
Clement

C:\Users\Clement\Desktop>pause
Press any key to continue . . .
```

"set" is used to set variables

a space after the "=" means it will include a space

surround with '%' to use a variable

```
set name=Clement
echo %name%
pause
```

```
C:\Users\Clement\Desktop>set name=Clement

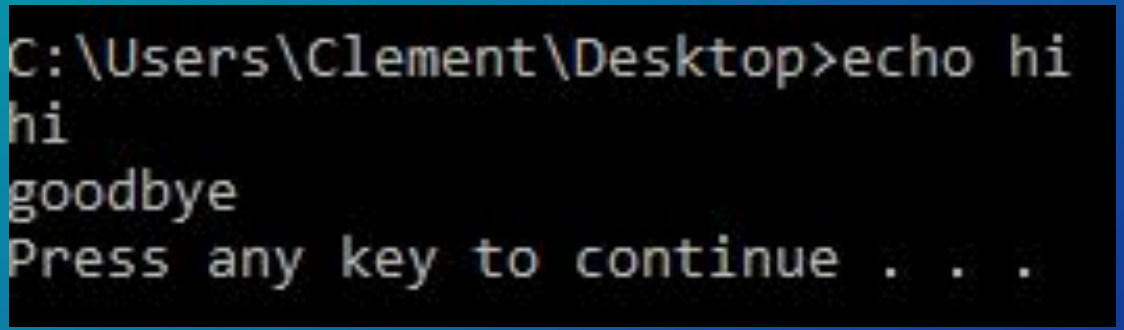
C:\Users\Clement\Desktop>echo Clement
Clement

C:\Users\Clement\Desktop>pause
Press any key to continue . . .
```

Batch substituted the variable in before executing the statement

You can use “@echo off” to suppress command inputs

```
echo hi
@echo off
echo goodbye
pause
```



Batch has the “GOTO” keyword to jump around

```
echo hi
GOTO someMarker
echo goodbye
:someMarker
pause
```

```
C:\Users\Clement\Desktop>echo hi
hi

C:\Users\Clement\Desktop>GOTO someMarker

C:\Users\Clement\Desktop>pause
Press any key to continue . . .
```

“goodbye” won’t be printed

The if statement uses () instead of the traditional {}

Supply a conditional - if it evaluates to true, the stuff inside runs

```
C:\Users\Clement>if %username% EQU Clement (echo hi)
hi

C:\Users\Clement>if %username% EQU clement (echo hi)

C:\Users\Clement>
```

Use these when comparing two things in batch

EQU
NEQ
LSS
LEQ
GTR
GEQ

Evaluate the following:

"1 EQU 3" is false

"2 LSS 3" is true

"3 LEQ 3" is true

SYNTAX:

for [parameter] %%parameter in (set) do command

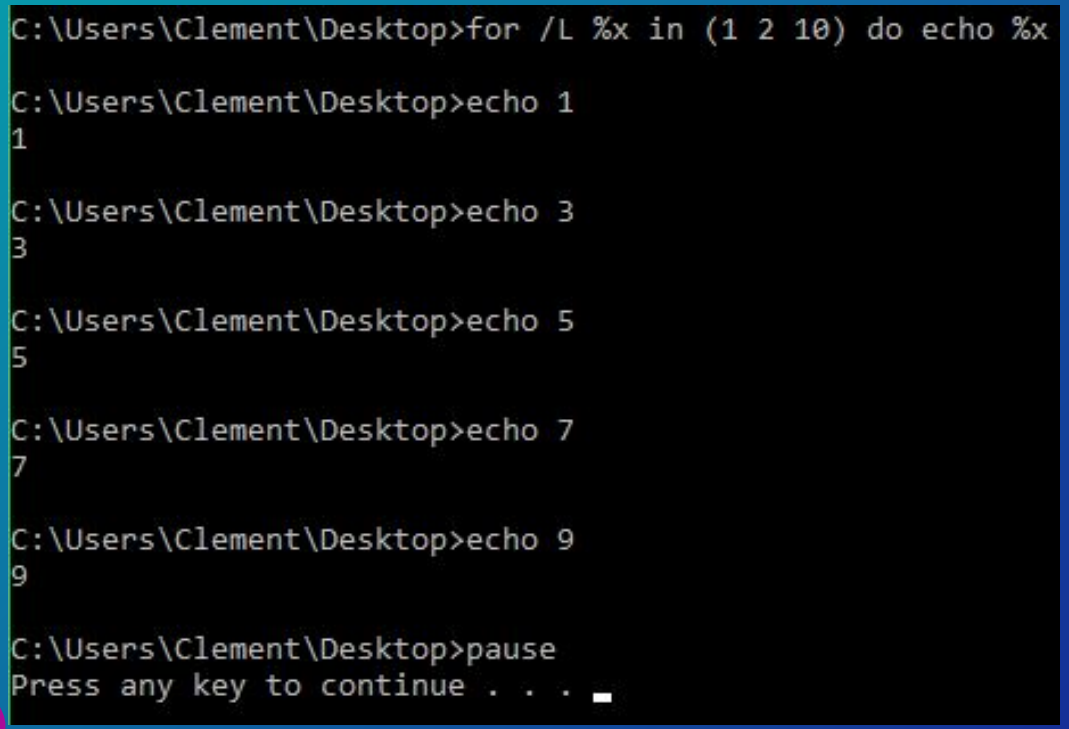
parameters for the “for” command are a bit confusing

check out ss64.com/nt/for.html

“for /L” interates numbers

for /L %%parameter in (start,step,end) do command

for /L %%x in (1,2,10) do echo %%x



"for /F" iterates in a list

for /F %%parameter in (filename) do command

for /F %%parameter in ('command') do command

```
@echo off
for /F %%u in (myusers.txt) do echo %%u
pause
```

myusers - Notepad

File Edit Format View Help

```
christo
clem
evan
jim
joseph
rahil
tim
```

C:\WINDOWS\system32\cmd.exe

```
christo
clem
evan
jim
joseph
rahil
tim
Press any key to continue . . .
```

SCRIPTING DESIGN *process*



Plan out what you want it to do

TEST, TEST, TEST

Write the code

Generic Scripting Example

Batching up a script



Planning - let's have our script:

- Auto open GUIs for you
- Set some settings

Coding - auto open GUIs

Opening GUIs can be done by simply supplying the command

appwiz.cpl
fsmgmt.msc
pause

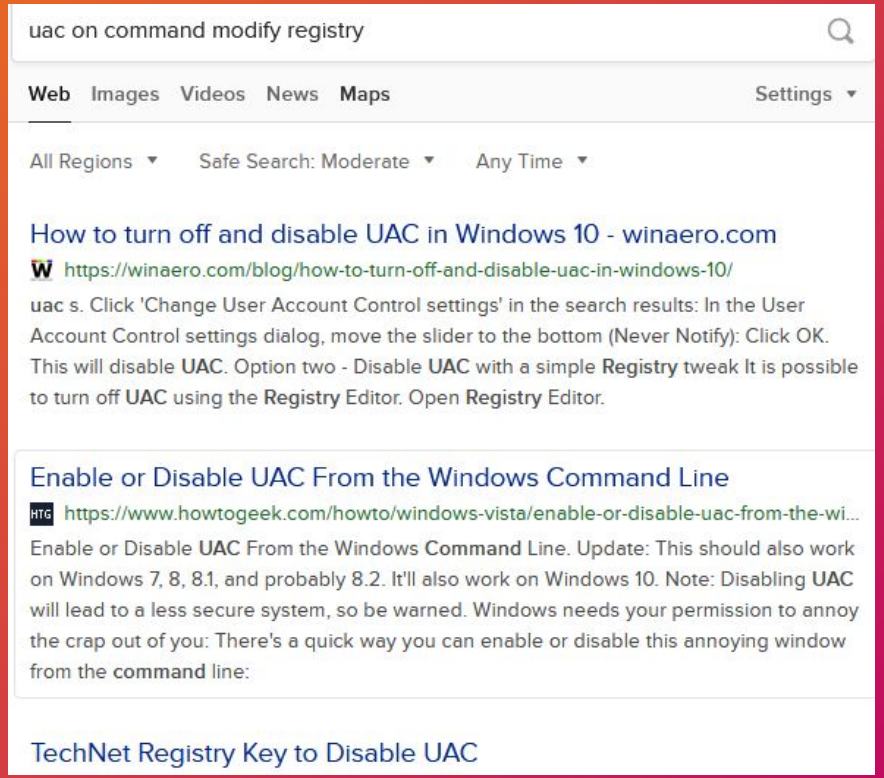
This will open the uninstall wizard and shares GUIs for you, then pause until you hit Enter

Batching up a Script

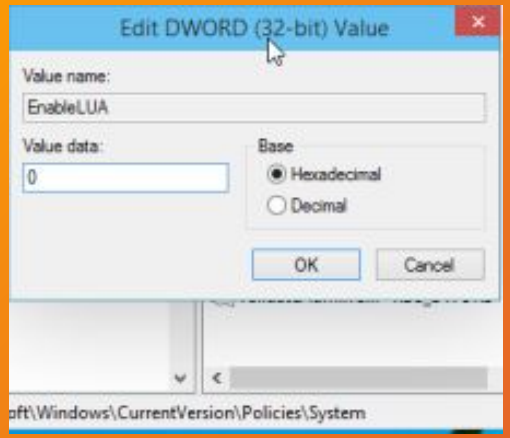
Coding - set some settings

Most settings can be set simply by modifying a registry key

Use the search engines to your advantage!



HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System



Enable UAC

```
C:\Windows\System32\cmd.exe /k %windir%\System32\reg.exe ADD
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v
EnableLUA /t REG_DWORD /d 1 /f
```

After you enable or disable UAC, you will have to reboot your computer for the changes to take effect.

Coding - set some settings

```
reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System
/v EnableLUA /t REG_DWORD /d 1 /f
reg add HKLM\SOFTWARE\Microsoft\...
reg add HKCU\SYSTEM\...
...
```

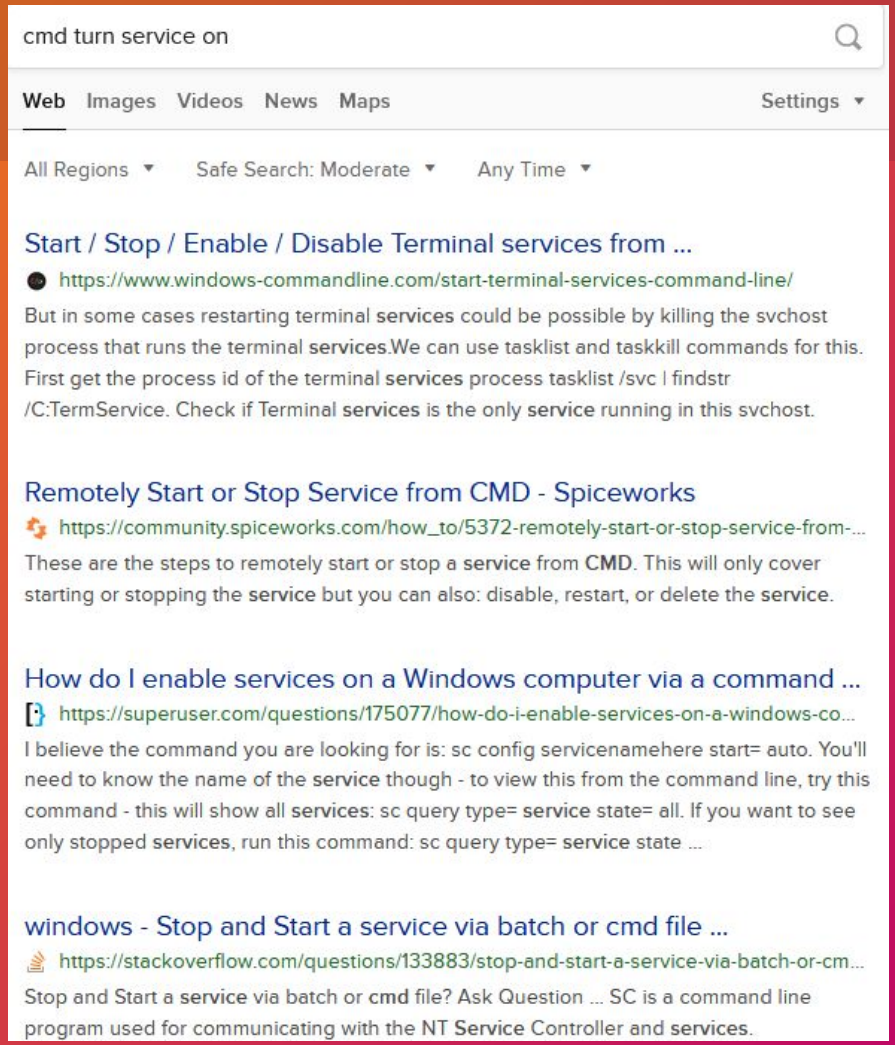
You can line up as many registry tweaks as you wish

This makes for effective scripting

Batching up a Script

Coding - set some settings

Some settings can also be set with a simple command



Testing

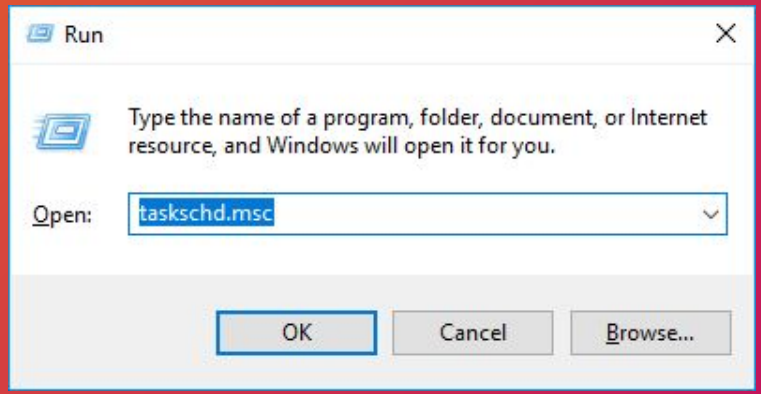
- Test several times with an image
- The last thing you want to do is break your image - make sure other tasks aren't affected

Task Scheduler

wow so cool

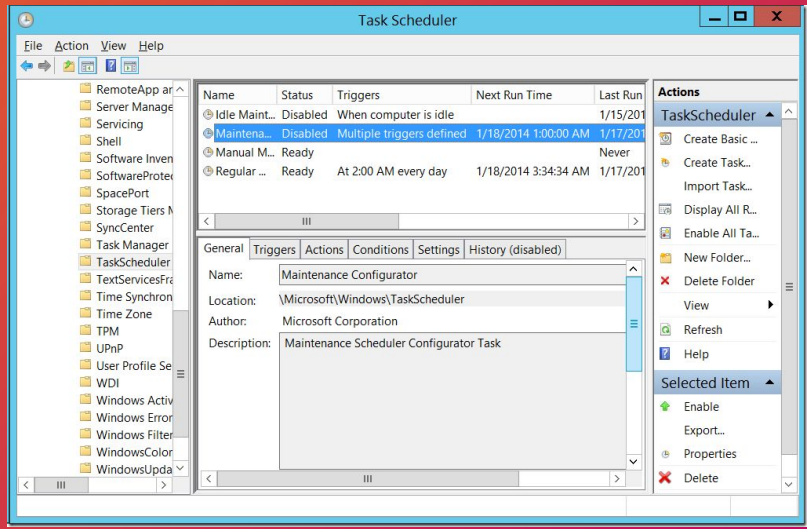
How to get to task Scheduler

- taskschd.msc
- mmc and add the snap in
- search task scheduler
- control panel -> system and security -> Administrative tools -> Schedule task



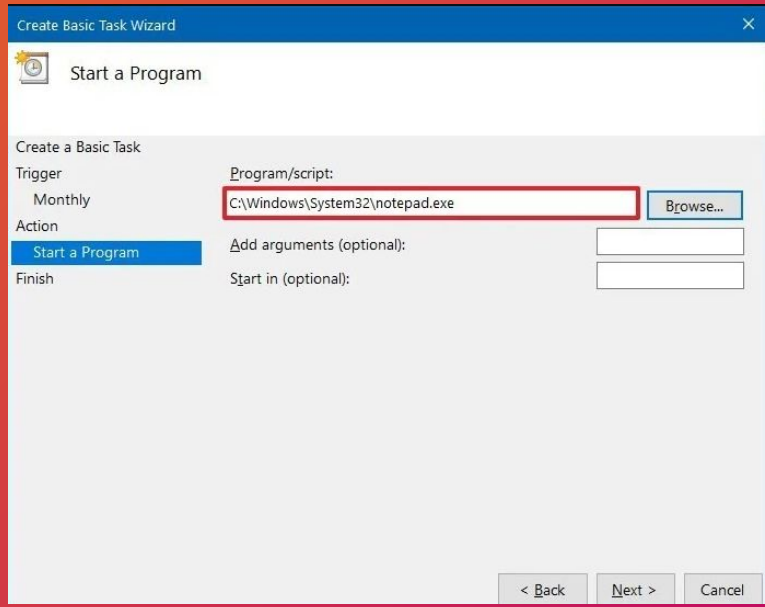
What is task scheduler

- where you schedule and view tasks
- can schedule programs to run after a trigger or timer
- runs .exe programs
- send an email
- display a message



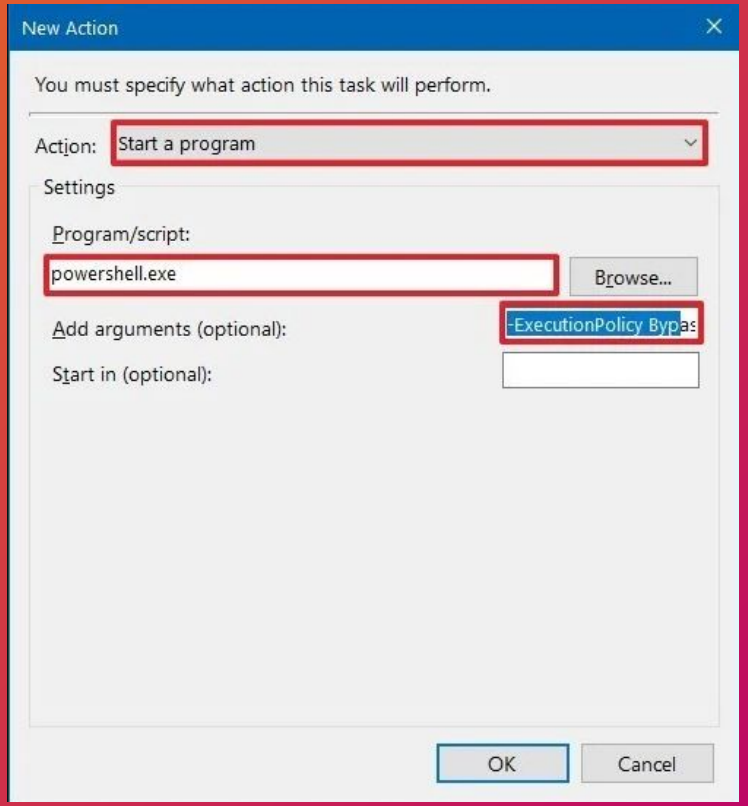
Basic Tasks

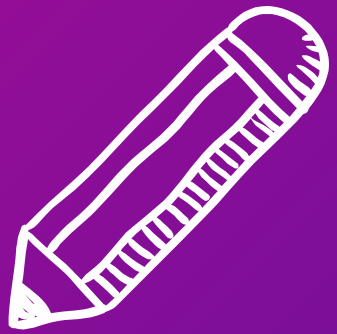
- right click and click create basic task
- can run .exe programs
  - by finding .exe file
  - like notepad.exe
- display message
  - will show message when triggered



Advanced Tasks

- adding powershell scripts to a task
- click on create task
  - schedule and name it
  - powershell.exe
  - put the pathing to your powershell script
  - set up conditions

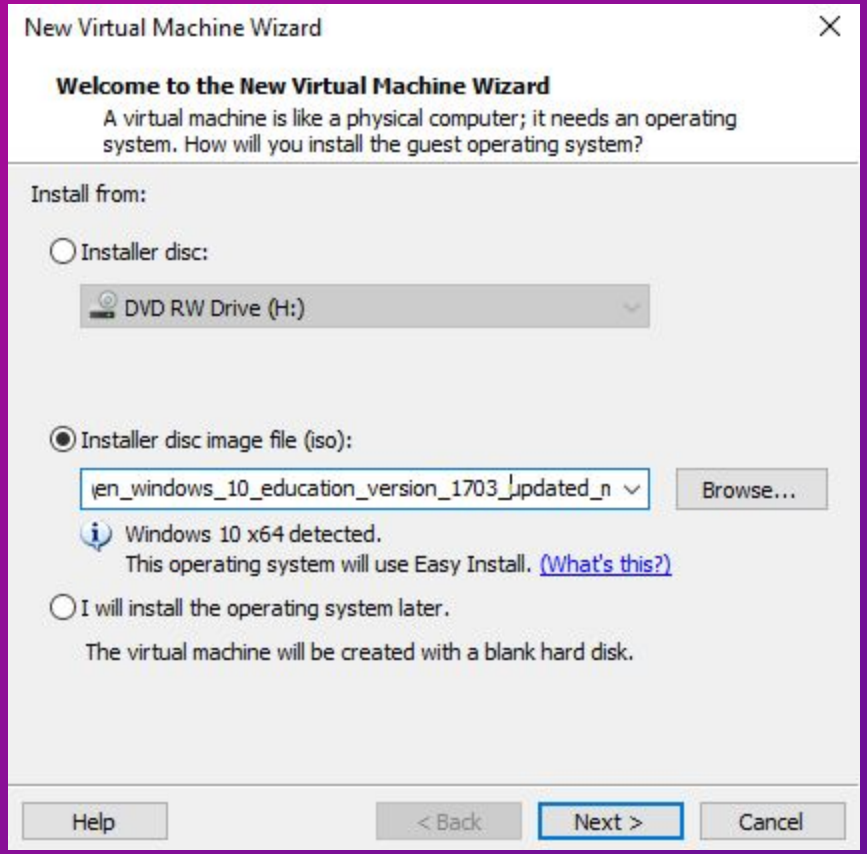


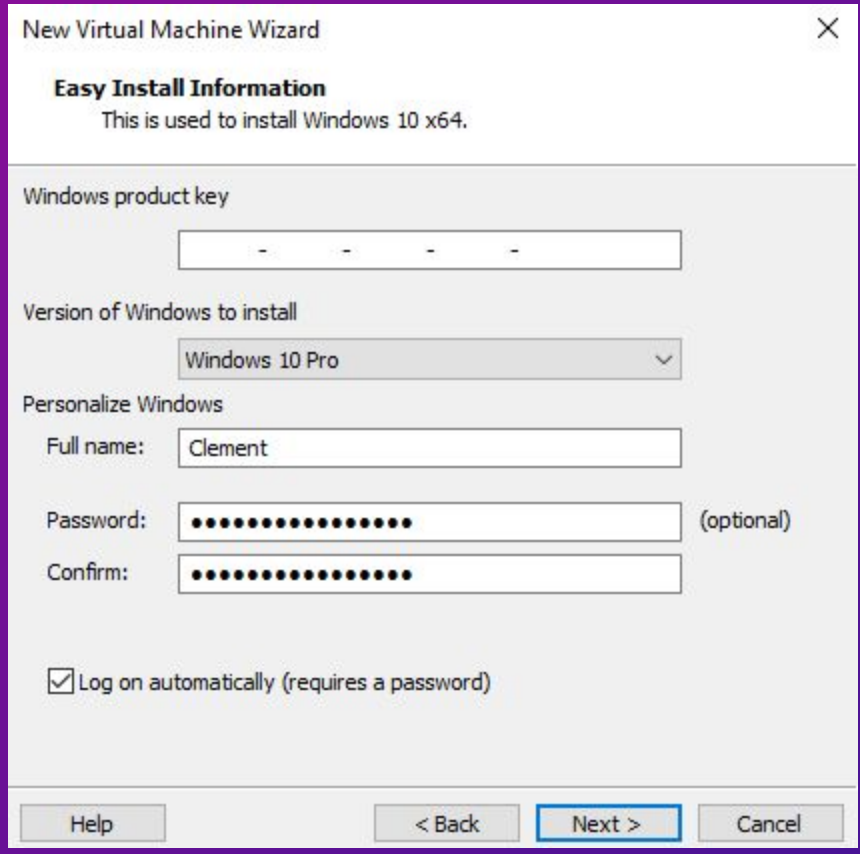


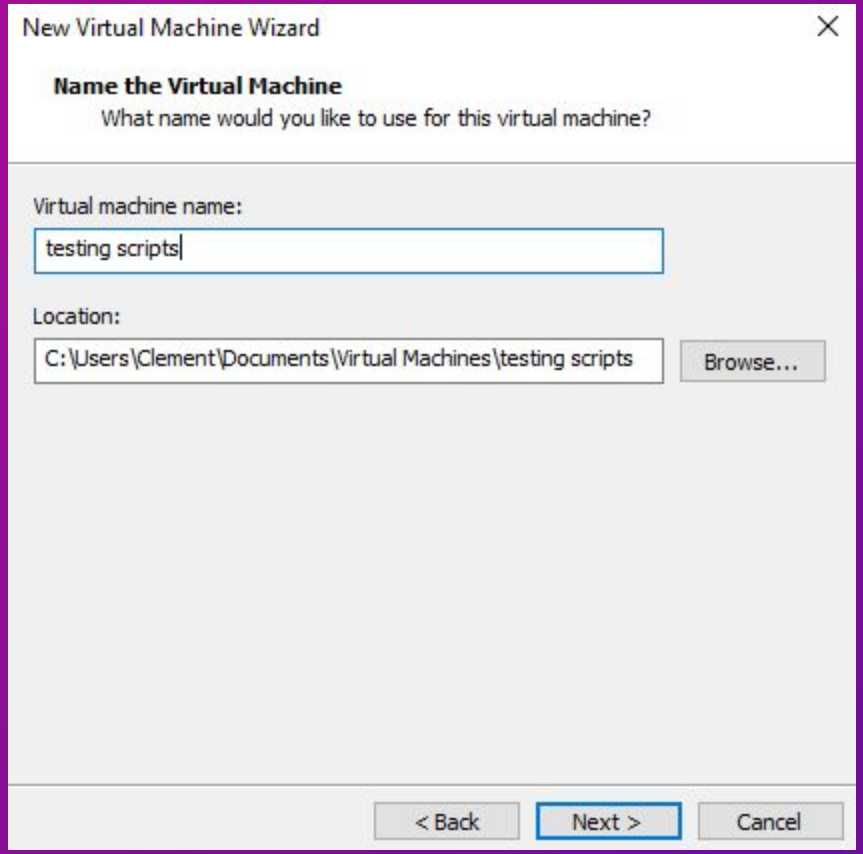


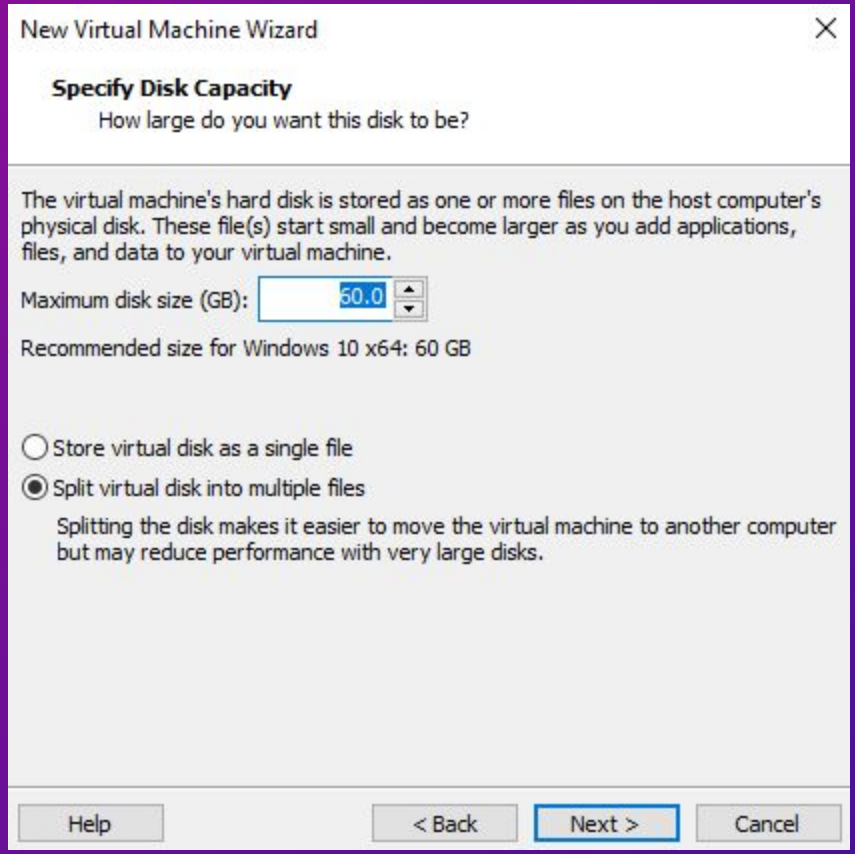
CREATING an *image*

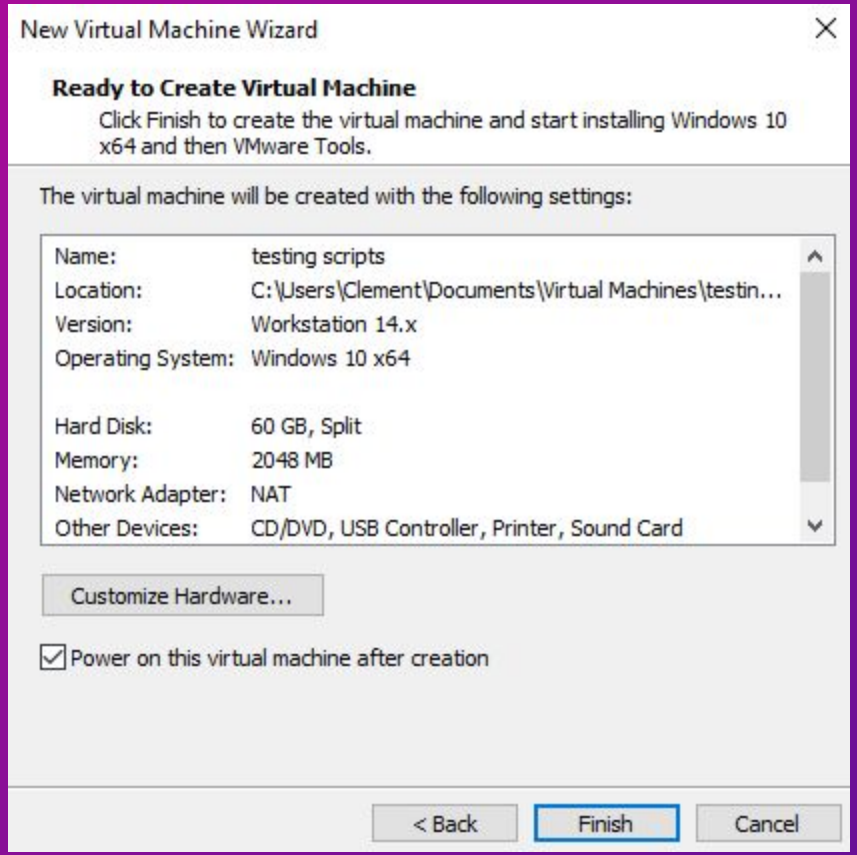
Use iso files to create a virtual machine

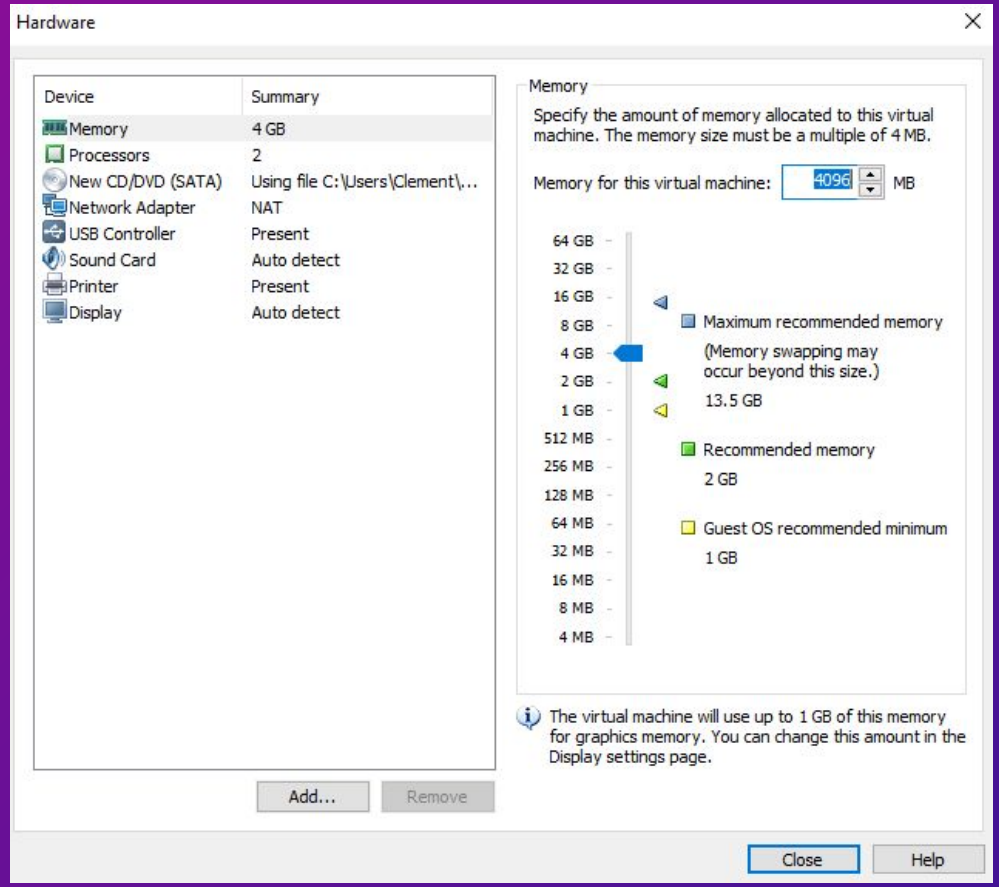

New Virtual Machine Wizard
Welcome to the New Virtual Machine Wizard
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
DVD RW Drive (H:)
Installer disc image file (iso):
en_windows_10_education_version_1703_updated_n
Browse...
Windows 10 x64 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Easy Install Information
This is used to install Windows 10 x64.
Windows product key
Version of Windows to install
Windows 10 Pro
Personalize Windows
Full name:
Clement
Password:
(optional)
Confirm:
Log on automatically (requires a password)
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
testing scripts
Location:
C:\Users\Clement\Documents\Virtual Machines\testing scripts
Browse...
< Back
Next >
Cancel


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
50.0
Recommended size for Windows 10 x64: 60 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Windows 10 x64 and then VMware Tools.
The virtual machine will be created with the following settings:
Name: testing scripts
Location: C:\Users\Clement\Documents\Virtual Machines\testin...
Version: Workstation 14.x
Operating System: Windows 10 x64
Hard Disk: 60 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: CD/DVD, USB Controller, Printer, Sound Card
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel


Hardware
Device
Summary
Memory
4 GB
Processors
2
New CD/DVD (SATA)
Using file C:\Users\Clement\...
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Printer
Present
Display
Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine:
4096
MB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.5 GB
Recommended memory
2 GB
Guest OS recommended minimum
1 GB
The virtual machine will use up to 1 GB of this memory for graphics memory. You can change this amount in the Display settings page.
Close
Help

The machine is now ready to boot and be set up

Follow through the instructions and wait for it to finish.


Script on!
ANY QUESTIONS?
You can find more help at:
Google
StackOverflow
and more!
HACKERMAN

These icons are editable shapes.

Examples:





Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash